Antyabha Rahman

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Education

University of New South Wales Jun 2026 – Present
Bachelor of Science (Honours) in Computer Science and Engineering

University of New South Wales Sep 2023 – Jun 2026
Bachelor of Science (Computer Science) WAM: DN

The University of British Columbia Aug 2025 – Dec 2025
Exchange Program Vancouver, Canada

Grants

- Rapid Grant, BlueDot Impact ($8,000) 2026

Publications

1. *MASK-OFF: Evaluating LLMs for omission of material fact*
   A. Rahman, Justin Michael Moeljono
   Trustworthy AI for Good Workshop @ NeurIPS (2026) [link]
2. *Probing the Origins of Reasoning Performance: Representational Quality for Mathematical Problem-Solving in RL vs. SFT Fine-Tuned Models*
   A. Rahman, A. Gurugubelli, O. Ankit, K. Zhu, A. Balwani
   XAI4Science Workshop @ AAAI (2026) [link]

Research Experience

Algoverse May 2025 – Jan 2026
Research Intern | GitHub Remote

- Designed an experiment comparing training stages, using linear probes on layerwise hidden states to test whether answer correctness is decodable in instruction-tuned vs. RL checkpoints of OLMo-3 and DeepSeek-Math on synthetic GSM8K-style problems.
- Implemented an inference pipeline using vLLM on an NVIDIA GH200 GPU for 24,000+ generations across 11 models for token-variability experiments.

Industry Experience

Children's Cancer Institute Feb 2026 – May 2026
Capstone Project Sydney, NSW

- Migrated and refactored an internal gene-scoring pipeline from pandas/MySQL to PySpark and Spark SQL on Azure Databricks, reducing end-to-end runtime from ∼4 hours to less than 2 minutes.
- Developed an Explainable Boosting Machine (EBM) model with a team of curators to predict reportable variants, enabling inspection of feature importance per case.

Service

- Trustworthy AI for Good Workshop @ NeurIPS, Reviewer (3 papers) 2026

Volunteering

Arc – UNSW Student Life Aug 2024 – Dec 2024
eReuse Project Coordinator Sydney, NSW

- Grew volunteer engagement by introducing structured Pair-Repair sessions and community-building events.
- Managed logistics for 150+ donated devices, fulfilling 95% of repair requests by creating an organisational system for parts and developing relationships with donors.